

7.2 最佳平方逼近问题求解方法

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7.2.1 最佳平方逼近问题求解初探

最佳平方逼近及其计算

对 $f(x) \in C[a, b]$ 及 $C[a, b]$ 中的一个子集

$$\varphi = \text{span}\{\varphi_0(x), \varphi_1(x), \dots, \varphi_n(x)\}$$

若存在 $S^*(x) \in \varphi$ ，使

$$\begin{aligned} \|f(x) - S^*(x)\|_2^2 &= \min_{S(x) \in \varphi} \|f(x) - S(x)\|_2^2 \\ &= \min_{S(x) \in \varphi} \int_a^b \rho(x) [f(x) - S(x)]^2 dx. \end{aligned} \quad (7.2.1)$$

则称 $S^*(x)$ 是 $f(x)$ 在子集 $\varphi \subset C[a, b]$ 中的最佳平方逼近函数. $S^*(x) = a_0^* \varphi_0(x) + \dots + a_n^* \varphi_n(x)$.

由 (7.2.1) 可知该问题等价于求多元函数

$$I(a_0, a_1, \text{L}, a_n) = \int_a^b \rho(x) \left[\sum_{j=0}^n a_j \varphi_j(x) - f(x) \right]^2 dx \quad (7.2.2)$$

的最小值.

$I(a_0, a_1, \text{L}, a_n)$ 是关于 a_0, a_1, L, a_n 的二次函数,

利用多元函数求极值的必要条件

$$\frac{\partial I}{\partial a_k} = 0 \quad (k = 0, 1, \text{L}, n),$$

即

$$\frac{\partial I}{\partial a_k} = 2 \int_a^b \rho(x) \left[\sum_{j=0}^n a_j \varphi_j(x) - f(x) \right] \varphi_k(x) dx$$
$$(k = 0, 1, \text{L}, n),$$

于是有

$$\sum_{j=0}^n (\varphi_k(x), \varphi_j(x)) a_j = (f(x), \varphi_k(x)) \quad (k = 0, 1, \dots, n). \quad (7.2.3)$$

这个关于 $a_0, a_1, \dots, a_n$ 的线性方程组，称为**法方程组**.

法方程组

$$\begin{pmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \text{L} & (\varphi_0, \varphi_n) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \text{L} & (\varphi_1, \varphi_n) \\ \text{L} & \text{L} & \text{L} & \text{L} \\ (\varphi_n, \varphi_0) & (\varphi_n, \varphi_1) & \text{L} & (\varphi_n, \varphi_n) \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \text{M} \\ a_n \end{pmatrix} = \begin{pmatrix} (f, \varphi_0) \\ (f, \varphi_1) \\ \text{M} \\ (f, \varphi_n) \end{pmatrix}$$

由于 $\varphi_0(x), \varphi_1(x), \text{L}, \varphi_n(x)$ 线性无关, 故

$$\det G(\varphi_0, \varphi_1, \text{L}, \varphi_n) \neq 0$$

于是方程组有唯一解 $a_k = a_k^* \quad (k = 0, 1, \text{L}, n),$

从而得到

$$S^*(x) = a_0^* \varphi_0(x) + \text{L} + a_n^* \varphi_n(x).$$

定理 设 X 为一个内积空间, $u_1, u_2, \dots, u_n \in X$,

矩阵

$$G = \begin{bmatrix} (u_1, u_1) & (u_2, u_1) & \dots & (u_n, u_1) \\ (u_1, u_2) & (u_2, u_2) & \dots & (u_n, u_2) \\ \vdots & \vdots & \ddots & \vdots \\ (u_1, u_n) & (u_2, u_n) & \dots & (u_n, u_n) \end{bmatrix} \quad (7.2.4)$$

称为格拉姆 (Gram) 矩阵.

则 G 非奇异的充分必要条件是 $u_1, u_2, \dots, u_n$ 线性无关.

若令 $\delta(x) = f(x) - S^*(x)$, 则平方误差为

$$\begin{aligned}\|\delta(x)\|_2^2 &= (f(x) - S^*(x), f(x) - S^*(x)) \\ &= (f(x), f(x)) - (S^*(x), f(x)) \\ &= \|f(x)\|_2^2 - \sum_{k=0}^n a_k^*(\varphi_k(x), f(x)).\end{aligned}\tag{7.2.5}$$

例 $f(x) = x^4$, $x \in [-1, 1]$ 。求 $P(x) \in H_2$ (不超过二次的多项式全体), 使 $\int_{-1}^1 [f(x) - P(x)]^2 dx$ 最小。

解: 设 $P(x) = a + bx + cx^2$, 即取 $\varphi_0(x) = 1$, $\varphi_1(x) = x$, $\varphi_2(x) = x^2$, $\rho(x) = 1$

$$\begin{cases} (\varphi_0, \varphi_0)a + (\varphi_0, \varphi_1)b + (\varphi_0, \varphi_2)c = (\varphi_0, f) \\ (\varphi_1, \varphi_0)a + (\varphi_1, \varphi_1)b + (\varphi_1, \varphi_2)c = (\varphi_1, f) \\ (\varphi_2, \varphi_0)a + (\varphi_2, \varphi_1)b + (\varphi_2, \varphi_2)c = (\varphi_2, f) \end{cases}$$

$$(\varphi_0, \varphi_0) = \int_{-1}^1 dx = 2, \quad (\varphi_0, \varphi_1) = (\varphi_1, \varphi_0) = \int_{-1}^1 x dx = 0,$$

$$(\varphi_1, \varphi_2) = (\varphi_2, \varphi_1) = \int_{-1}^1 x^3 dx = 0,$$

$$(\varphi_1, \varphi_1) = (\varphi_0, \varphi_2) = (\varphi_2, \varphi_0) = \int_{-1}^1 x^2 dx = \frac{2}{3}, \quad (\varphi_2, \varphi_2) = \int_{-1}^1 x^4 dx = \frac{2}{5},$$

$$(\varphi_0, f) = \int_{-1}^1 x^4 dx = \frac{2}{5}, \quad (\varphi_1, f) = \int_{-1}^1 x^5 dx = 0, \quad (\varphi_2, f) = \int_{-1}^1 x^6 dx = \frac{2}{7}$$

$$\begin{cases} 2a + \frac{2}{3}c = \frac{2}{5} \\ \frac{2}{3}b = 0 \\ \frac{2}{3}a + \frac{2}{5}c = \frac{2}{7} \end{cases}$$

$$b = 0, \quad a = -\frac{3}{35}, \quad c = \frac{6}{7}$$

$$P(x) = -\frac{3}{35} + \frac{6}{7}x^2$$

误差:

$$\int_{-1}^1 (x^4 - P(x))^2 dx = 0.012$$

7.2.2 用正交函数族解最佳平方逼近问题

设 $f(x) \in C[a, b]$, $\varphi = \text{span}\{\varphi_0(x), \varphi_1(x), \dots, \varphi_n(x)\}$,
若 $\varphi_0(x), \varphi_1(x), \dots, \varphi_n(x)$ 是正交函数族,

则

$$(\varphi_i(x), \varphi_j(x)) = 0, \quad i \neq j$$

而

$$(\varphi_j(x), \varphi_j(x)) > 0$$

考察法方程组

$$\begin{pmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \text{L} & (\varphi_0, \varphi_n) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \text{L} & (\varphi_1, \varphi_n) \\ \text{L} & \text{L} & \text{L} & \text{L} \\ (\varphi_n, \varphi_0) & (\varphi_n, \varphi_1) & \text{L} & (\varphi_n, \varphi_n) \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \text{M} \\ a_n \end{pmatrix} = \begin{pmatrix} (f, \varphi_0) \\ (f, \varphi_1) \\ \text{M} \\ (f, \varphi_n) \end{pmatrix}$$

故法方程组的系数矩阵，为非奇异对角阵。

法方程组的解为

$$a_k^* = (f(x), \varphi_k(x)) / (\varphi_k(k), \varphi_k(x)) \quad (7.2.6)$$
$$(k = 0, 1, \dots, n).$$

于是 $f(x) \in C[a, b]$ 在 φ 中的最佳平方逼近函数为

$$S^*(x) = \sum_{k=0}^n \frac{(f(x), \varphi_k(x))}{\|\varphi_k(x)\|_2^2} \varphi_k(x). \quad (7.2.7)$$

误差为

$$\begin{aligned}\|\delta_n(x)\|_2 &= \|f(x) - S_n^*(x)\|_2 \\ &= \left(\|f(x)\|_2^2 - \sum_{k=0}^n \left[\frac{(f(x), \varphi_k(x))}{\|\varphi_k(x)\|_2} \right]^2 \right)^{\frac{1}{2}}. \quad (7.2.8) \\ &= \left(\|f(x)\|_2^2 - \sum_{k=1}^n (a_k^* \|\varphi_k(x)\|_2)^2 \right)^{\frac{1}{2}}.\end{aligned}$$

- End.